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28 July 1937

MILITARY SPECIFICATION

CIRCLE, AIMING, M1

This specification consists of this cover sheet, and Specification 51-71-25, dated 28 July 1937, attached hereto, without modification.

When a request for this specification is received by a supplying activity it will be necessary to attach this cover sheet to the pertinent specification before issue.

Copies of specifications required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.

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U. S. ARMY
SPECIFICATION

No. 51-71-25

JULY 28, 1937

CIRCLE, AIMING, M1

A. APPLICABLE SPECIFICATIONS.

A–1. Ordnance Department Drawing Class 22, Division 249, Drawing 1, titled "Circle, Aiming, M1, List of Drawings" lists all drawings and specifications necessary to complete the manufacture of this instrument.

A–2. All drawings and specifications listed on the above drawing, of the issue in effect on the date of invitation for bids, form part of this specification.

B. TYPE.

B–1. This specification covers one type of instrument known as Circle, Aiming, M1, complete with equipment except as stated in par. B–2.

B–2. The Tripod, M5, shown on the drawings as forming part of the complete instrument, is not covered by this specification.

C. MATERIAL AND WORKMANSHIP.

C–1. Material and workmanship shall be in accordance with drawings, general specifications, and material specifications forming parts of this specification.

C–2. Components and material of commercial nature, for which no requirements are contained herein, or for which no specifications are listed on the list of specifications drawing, shall be of the type most suitable for the purpose, and shall be of the best commercial quality unless a lower grade is specified.

C–3. In order to obtain the accuracy and performance required in section E, special attention is necessary to the following:

C–3a. *Cell, Reticle (A–39585).—*

Cell, Lens (A–39587).—

Cell, Objective (A39588).—Concentricity of bored diameter for optical elements with outside centering diameter.

C–3b. *Bracket (A–43889).*—(1) Parallelism of axis of trunnion hole with bottom attaching surface.

(2) Perpendicularity of faces of hole with axis of hole.

C–3c. *Worm (A–39608 and B–129650).*—(1) Concentricity of pitch circle of worm threads with bearing diameters.

(2) Uniformity of lead and smoothness of threads.

C–3d. *Plate (A–39622 and A–39619).*—Concentricity of micrometer centering diameter with hole.

C–3e. *Trunnion (B–129644).*—Concentricity of diameter that fits telescope with large and small bearing diameters that fit body.

C–3f. *Body, telescope (D–28553).*—(1) Parallelism of seat planes for covers and prisms with each other and with the axis of trunnion bearings.

(2) Perpendicularity of axis of level vial hole with plane of the seats for the covers.

C–3g. *Wheel, worm (C–44750).*—(1) Concentricity of pitch circle of worm teeth, tapered diameters, and graduated diameter with center hole.

(2) Uniformity of lead and smoothness of teeth.

(3) Lapping the worm wheel to the body (D–28553) in a manner that will insure a bearing on both the tapered diameter and shoulder.

C–3h. *Housing, worm (C–44752).*—(1) Concentricity of cylindrical bearing surface for worm wheel with hole for spindle.

(2) Perpendicularity of axis of worm hole with axis of spindle hole.

C–3i. *Body (D–28553).*—(1) Concentricity of tapered hole for needle pivot with tapered bearing that fits the worm wheel.

(2) Perpendicularity of attaching surface for bracket, axis of worm hole, and axis of hole for trunnion bearing to axis of tapered bearing.

(3) Machining the circular track for worm shoe and the worm plunger hole parallel to each other and so that the longitudinal axes of the track and plunger hole intersect the axis of the worm hole.

C–3j. Fitting body to worm wheel and worm wheel to worm housing and securing them by means of washer and special screw to give a smooth revolving fit free from perceptible shake or lift in operation.

C–3k. Machining and fitting the reticle holder (A–43928) and the end piece (C–44995) to insure parallelism of the reticle line with respect to the vertical axis of the instrument.

D. GENERAL REQUIREMENTS.

D–1. See U. S. Army Specification No. 51–7, "General Specification Governing the Manufacture and Inspection of Fire Control Instruments."

E. DETAIL REQUIREMENTS.

E–1. Interchangeability shall be as defined in the paragraph on interchangeability in specification No. 51–7.

E–2. *Accuracy and adjustment.—*

E–2a. All parts shall be assembled as per assembled view drawings.

E–2b. All rotating and moving parts shall function properly without irregularities, undue friction or looseness.

E–2c. The stop mechanism of the telescope elevation screw shall be adjusted to restrain the telescope from striking the compass window and shall allow at least 8° movement of the telescope above and 8° below the horizontal.

E–2d. With the axes of the two azimuth worms approximately parallel to each other and on opposite sides of the vertical axis of the instrument, and with zero index arrow of the scale (B–135656) set to coincide with the zero on the azimuth scale, the micrometers shall be set to indicate zero.

E–2e. The illuminating mechanism shall function as intended to illuminate the figures and graduations. When tested in a dark room, the figures and graduations of the reticle shall appear clearly defined.

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E–2f. The reticle shall be in sharp focus of the eyepiece when the focusing sleeve is at the middle of its movement within the tolerance of .05″, measured on the circumference of the eyepiece adapter, when the reticle is observed through a magnifying telescope of at least three power.

E–2g. There shall be no parallax between the image and the reticle lines when a target such as a corner of a building, pole, or stack is viewed at a distance of 75 to 85 yards.

E–2h. The telescope shall produce an image practically free from distortion, astigmatism, curvature of field, coma and chromatic, and spherical aberration for a radial distance of 40 mils from the center of the field of view. Images showing any of these defects in the specified area shall be cause for rejection of the telescope when the defect is observable through the telescope and is judged objectionable by the inspector.

E–2i. With the instrument mounted on a truly vertical spindle and with the optical axis of the telescope horizontal, the instrument shall have the following adjustment:

(1) The tubular level bubble shall be central with respect to the level vial graduations within the tolerance of 0.33 mil.

(2) The circular level bubble shall be central within the limits of the etched circle of the vial.

(3) The horizontal and vertical crosslines shall be horizontal and vertical respectively within a tolerance of 0.5 mil when checked against crosslines of a testing target.

(4) The line of sight of the emergent optical axis shall be parallel to a vertical plane passing through the magnetic needle pivot and the two opposite magnetic needle indexes within the tolerance of 0.5 mil (1′ 41″).

E–2j. With the instrument mounted on a truly vertical spindle on a leveled azimuth testing fixture, the line of sight shall not deviate up or down from a point on a vertical line of a target by more than the tolerances given below when the telescope is rotated to any position in 360° and is brought back to the target by means of the azimuth testing fixture:

(1) When rotated by the upper azimuth mechanism, 1 mil (3′ 22″).

(2) When rotated by the lower azimuth mechanism 0.5 mil (1′ 41″).

(3) When the whole instrument is rotated about the vertical spindle, 0.5 mil (1′ 41″).

(4) When rotated by any combination of the three above means, 1.5 mils (5′ 3″).

E–2k. In several clamping and unclamping tests of the instrument to the vertical spindle the leveling of the instrument shall not vary more than 0.5 mil (1′ 41″).

E–2l. With the instrument mounted as in paragraph E–2j, the optical line of sight shall follow a plumb line within the tolerance of 0.5 mil (1′ 41″) when the telescope is rotated through the full vertical movement at any azimuth setting.

E–2m. In reversing the azimuth motion of the telescope by means of either the upper or lower azimuth worm knob, the displacement between the line of site and a horizontal line on a target shall not exceed 0.3 mil (1′ 7″).

E–2n. The 60 mil reticle graduations of the mil and stadia scales shall coincide with the 60-mil graduations of the mil scale of an accurate testing target within the tolerance of 1.5 mils when the zero vertical and horizontal lines of the reticle and target are in coincidence. The reticle graduations shall appear uniformly spaced.

E–2o. Backlash in the azimuth mechanism shall not exceed 0.5 mil (1′ 41″). The error in any azimuth reading shall not exceed 1.5 mils including backlash.

E–2p. The magnetic needle shall be perfectly balanced, well magnetized, and mounted to the jewel housing so that the knife edge end, the graduated line, and the center of the pivot bearing are in alinement. North seeking end must point to the magnetic north when released and shall swing with a uniform motion with no appreciable dip. In several tests the needle shall come to rest each time in the same direction; namely, the magnetic north. The magnetic needle mechanism shall allow the needle to function when the instrument is 5° 30′ out of level.

E–2q. The half-period shall be less than four seconds. The half-period is defined as the time interval between successive transits of the north end of the needle through its position of equilibrium.

E–3. The instrument, exclusive of the magnetic needle and pivot, shall be free from any painting, plating, or metal which contains more than 0.25 of 1 percent of iron.

E–4. The interior of the telescope and compass needle compartment shall be sealed against the entrance of water and moisture.

E–5. Lubrication shall be in accordance with specification U. S. Army No. 51–7.

E–6. Painting and finishing shall be in accordance with drawing and U. S. Army Specification No. 51–70–1, "Painting and Finishing Fire Control Instruments."

E–7. Manufacture of extra parts shall be in accordance with this specification.

E–8. Serial numbers will be furnished by the procuring office, and shall be inscribed in the manner shown on the drawings.

F. METHODS OF INSPECTION, TESTS, ETC.

F–1. Each instrument and component parts thereof, including extra parts, shall be inspected during process of manufacture and assembly, as provided by U. S. Army Specification No. 51–7, "General Specification Governing the Manufacture and Inspection of Fire Control Instruments."

F–2. Each completed instrument shall be tested for conformance with the requirements in section E.

F–3. One instrument out of each ten, selected by the inspector, shall be given the rain test, as prescribed in U. S. Army Specification No. 51–7. The requirements of this test shall apply to the compass needle compartment as well as to the interior of the telescope.

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F–4. Each instrument shall be inspected for accuracy of azimuth readings at at least 16 different settings throughout the 6,400 mils of azimuth. The settings shall be selected by the inspector.

G. PACKING AND MARKING.

G–1. *Packing.*—

G–1a. When the Aiming Circles are procured complete with equipment they shall be assembled to the tripod, placed in the carrying case and packed for shipment in wooden boxes in quantities not exceeding six per box, unless otherwise specified by the procuring office.

G–1b. When the Aiming Circles are procured without equipment they shall be packed in quantities specified by the procuring office.

G–1c. All packing shall be done in substantial wooden packing boxes with tongue and groove or lap joints, so constructed as to insure acceptance by common or other carriers for safe transportation at lowest rate to point of delivery.

G–2. *Marking.*—

G–2a. Marking shall be in accordance with U. S. Army Specification No. 100–2, "Standard Specification for Marking Shipments."

H. NOTES.

H–1. The use of this specification, whenever applicable, is mandatory on all procuring agencies of the Army.

NOTICE.—Whenever Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States thereby incurs no responsibility or any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded, by implication or otherwise, as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may be in any way related thereto.

NOTE.—Copies of this specification may be obtained from the Office, Chief of Ordnance, Washington, D. C.

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